

Roll No. :

END SEMESTER EXAMINATION December 2024

Name of the Program: **B.Tech**

Semester: VII

Name of the Course: Advanced Computer Architecture

Course Code: TCS 704

Time : 3 Hours

Maximum Marks : 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Discuss the concepts of coherence and locality properties in memory hierarchy design.	CO1
(b)	Using Amdahl's Law, determine the speedup for a program where 60% of the code is parallelized and the remaining 40% is sequential, with 4 processors.	CO3
(c)	Design an optimization strategy to improve cache performance in a given system. Discuss its impact on coherence and locality.	CO2
Q2	(20Marks)	
(a)	Describe the five-stage RISC pipeline and highlight common performance issues in pipelining.	CO5
(b)	A system has a cache hit rate of 95% and a memory access time of 100 ns. If the cache access time is 10 ns, calculate the average memory access time.	CO4
(c)	Evaluate the trade-offs between message-passing and shared-memory architectures in a multiprocessor system for a real-world application like database management.	CO4
Q3	(20Marks)	
(a)	Explain instruction-level parallelism (ILP) and the significance of RAW and WAW dependencies.	CO5
(b)	A processor has a clock rate of 2 GHz. It executes a program with 5 billion instructions, where the CPI (cycles per instruction) is 1.5. Calculate the execution time.	CO1
(c)	Propose a solution to mitigate pipeline hazards in a RISC processor. Analyze the effectiveness of your approach.	CO2

Q4	(20 Marks)	
(a)	Differentiate between centralized shared-memory and distributed shared-memory architectures.	CO2
(b)	For a RISC processor, determine the pipeline performance assuming a stall occurs every 4 cycles due to hazards. Given a clock cycle time of 2 ns, calculate the effective instruction execution rate (in MIPS).	CO5
(c)	Suppose that we want to enhance the processor used for Web serving. The new processor is 10 times faster on computation in the Web serving application than the original processor. Assuming that the original processor is busy with computation 40% of the time and is waiting for I/O 60% of the time, what is the overall speedup gained by incorporating the enhancement?	CO4
Q5	(20 Marks)	
(a)	What is branch prediction? Explain the performance metrics of computers.	CO1
(b)	A memory system has a cache size of 64 KB and a block size of 16 bytes. Calculate the number of cache blocks.	CO4
(c)	Explain classification of parallel computers based on i. Grain Size ii. Instruction and Data Stream.	CO3